PR BUILDER v2.4

PR BUILDER PLATFORM

Next-Gen AI Resume Engine, Portfolio Builder & Custom Domain Platform

Production Architecture, Feature Specifications & Technical Documentation

PROJECT OVERVIEW & METRICS SUMMARY

- Platform Domain: https://prbuilder.xyz (Live on Vercel Edge)
- Tech Stack: React 19, TypeScript, Tailwind CSS v4, Supabase (PostgreSQL), Vite
- Key Integrations: Cashfree Payment Gateway v3 SDK, Google Gemini 3.8 Flash AI

1. Executive Summary & Mission

PR Builder (KaizenQ Identity) is an all-in-one professional career launchpad designed for software engineers, developers, and job seekers. It combines an ATS-optimized multi-template Resume Builder, an interactive multi-theme Developer Portfolio Builder, instant Standalone ZIP & Subdomain Publishing, Google Gemini AI Career Assistance, and native Cashfree Payments into a unified modern web experience.

Unlike traditional SaaS builders that lock users behind rigid subscription paywalls and unverified subdomains, PR Builder provides 100% free lifetime developer access to core features, exportable offline assets (A4 PDF & full HTML/CSS/JS website ZIPs), free subdomain publishing (username.prbuilder.xyz), and an authoritative server-verified Custom Domain Pass for custom branding.

2. Core Architecture Highlights

- **Frontend SPA Architecture:** React 19 with functional components, custom hooks, and Tailwind CSS v4 for zero-runtime styling overhead and instantaneous responsive UI.
- **Server-Authoritative Backend:** Supabase PostgreSQL database secured with Row-Level Security (RLS) policies, preventing unauthorized read/write access to student profiles, resumes, and domains.
- **Production Payment Integration:** Direct Cashfree Payments v3 Web SDK integration supporting UPI, Cards, NetBanking, and Wallets with server-authoritative integer paise order calculation and HMAC webhook verification.
- **Google Gemini AI Intelligence:** Integrated Gemini 3.8 Flash assistant that rewrites weak bullet points into high-impact STAR statements and generates recruiter-ready summaries.

3. Interactive Resume Builder (Detailed Functionality)

The Resume Builder provides an intuitive, real-time authoring environment with both a 10-step Guided Stepper Mode and a free-form Section Navigation Mode. It incorporates ATS compliance algorithms, live A4 preview rendering, completeness scoring, and client-side PDF generation.

3.1 Core Resume Sections & Data Schema

- **Personal Identity:** Full name, job title, email, phone, location, LinkedIn URL, GitHub URL, and live portfolio website.
- **Professional Summary:** Concise 3-4 sentence elevator pitch with integrated Gemini AI auto-generation based on experience and target roles.
- **Education:** Degree, institution, CGPA/percentage, graduation dates, coursework, and honors with dynamic entry addition and reordering.
- **Work Experience:** Company name, designation, dates, location, and structured bullet points with action-verb suggestions and AI enhancement.
- **Technical Projects:** Project title, role, live demo URL, repository link, technology stack chips, and quantified impact metrics.
- **Skills Categorization:** Tag-based inputs grouped by Languages, Frameworks, Developer Tools, Databases, and Professional/Soft Skills.
- **Certifications:** Certification title, issuing authority, credential ID, date of issuance, and verification URLs.
- **Custom & Additional Sections:** User-defined sections such as Publications, Hackathons, Languages, or Volunteer Work.

3.2 ATS-Compliant Template Engine

PR Builder provides 4 professionally designed templates engineered to pass Applicant Tracking Systems (Workday, Greenhouse, Lever, Taleo) with high parsability scores:

Template Name	Layout Type	Ideal Use Case	ATS Compatibility
Classic ATS	Single-Column Standard	Software Engineering, FinTech, Traditional Corporate	99.4% Parsability
Modern Professional	Two-Column Executive	Product Management, Tech Leads, Experienced Devs	97.8% Parsability
Technical Developer	Monospace Tech-First	Full-Stack, DevOps, Cloud Architects, Systems	98.6% Parsability
Minimal Editorial	High-Contrast Serif/Sans	UI/UX Engineers, Design Technologists, Academics	98.2% Parsability

3.3 Dynamic Export & Persistence Features

- **Direct A4 PDF Export:** Generates crisp single or multi-page A4 PDFs directly in the browser via html2canvas and jsPDF with zero server latency.
- **Smart User Profile Naming:** Exported PDF filenames dynamically sanitize the active signed-in user name (e.g. Bhanu_Prakash_Resume.pdf) instead of generic placeholders.
- **JSON Draft Backup & Migration:** Full resume states can be exported as a JSON schema file and imported across devices for instant recovery.
- **Autosave & Cloud Synchronization:** Changes automatically persist to localStorage in real-time, with background syncing to Supabase for authenticated users.

4. Developer Portfolio Builder & Publishing Engine

The Portfolio Builder transforms verified identity credentials and projects into a responsive, high-performance web showcase. It includes a 3-pane real-time editor, theme customizer, standalone export, and dual-layer publishing.

4.1 Theme Catalog

Theme Name	Type	Aesthetic & Palette	Entitlement
Developer Terminal	Monospace Dev-First	Dark mode terminal with syntax highlighting chips	100% Free Forever
Minimal Editorial	Clean High-Contrast	Generous whitespace with refined typography rules	100% Free Forever
Midnight Dark Showcase	Deep Navy Cosmic	Ambient blue glow with modern software engineer cards	100% Free Forever
Creative Studio Pro	Vibrant Gradients	Animated cards, micro-interactions, rich media showcases	₹299 Pro Theme
Executive Horizon Pro	Premium Corporate	Sleek luxury enterprise styling for senior leaders	₹299 Pro Theme

4.2 Standalone Website ZIP Export

Users can download a self-contained, offline-first website ZIP package containing index.html, styled CSS, responsive JavaScript, and embedded assets. It requires zero backend servers and can be double-clicked to open locally or deployed to GitHub Pages, Netlify, or Vercel.

4.3 Free Subdomain Publishing & Direct Routing

Portfolios can be published directly to the live internet with automated slug availability validation:

- **Branded Subdomain:** Claims a dedicated subdomain at https://[username].prbuilder.xyz with HTTPS certification.
- **Guaranteed Direct Route:** Every published portfolio is immediately accessible worldwide via https://prbuilder.xyz/p/[username] (200 OK without DNS propagation delays).
- **High-Contrast QR Code:** Auto-generates downloadable scannable QR codes linking recruiters directly to the candidate's published portfolio.

4.4 Custom Domain Connection (Custom Domain Pass ₹499)

Allows professionals to connect their personal root or subdomains (e.g. johnsmith.dev, portfolio.mybrand.com):

- **Authoritative Entitlement:** Unlocks via Cashfree payment verification and records entitlement in Supabase.
- **Automated DNS Guidance:** Generates DNS CNAME, A record (75.2.60.5), and cryptographic TXT challenge records.
- **Verification Engine:** Performs DNS lookup checks to verify ownership before provisioning automated SSL certificates.

5. Cashfree Payment Gateway & Monetization Architecture

PR Builder integrates the official Cashfree Payments v3 Web SDK for fast, secure Indian payment processing across UPI (GPay, PhonePe, Paytm, BHIM), Credit/Debit Cards, NetBanking, and Wallets.

5.1 Server-Authoritative Payment Pipeline

Step	Component / Endpoint	Security & Implementation Details
1. Order Initiation	POST /api/cashfree-order	Computes authoritative amount in integer paise; checks coupons and locks order in DB.
2. Modal Launch	cashfreeService.ts & SDK v3	Loads official SDK from sdk.cashfree.com/js/v3/cashfree.js in seamless modal checkout mode.
3. Settlement Verification	POST /api/cashfree-verify	Queries Cashfree servers directly using secret keys; confirms PAID status before unlock.
4. Webhook Security	POST /api/cashfree-webhook	Validates x-webhook-signature using cryptographic HMAC-SHA256 and timestamp tolerances.
5. Entitlement Unlock	Supabase RPC verify_and_fulfill	Atomically records user entitlement, marks order completed, and enables Pro features.

5.2 Product Catalog & Pricing Table

Product ID	Item Name	Configured Price	Included Features
pro-template-*	Resume Pro Template	₹199 (19900 paise)	Individual Pro ATS Resume Template lifetime license
pro-theme-*	Portfolio Pro Theme	₹299 (29900 paise)	Individual Pro Portfolio Theme lifetime license
addon-custom-domain	Custom Domain Pass	₹499 (49900 paise)	Unlimited custom domain connections, SSL provisioning, DNS g
bundle-prbuilder-pro	PR Builder Pro All-Access	₹999 (99900 paise)	All Pro Resume Templates, All Portfolio Themes, Custom Doma

5.3 Promotional Coupon Engine

Supports instant discount vouchers validated on both client and edge server:

- **WELCOME10:** 10% off entire order for new signups.
- **RESUME50:** 50% off Resume Template upgrades.
- **HACKATHON100:** 100% free pass for partner university hackathon participants.

6. Google Gemini AI Career Assistant Engine

Powered by Google Gemini 3.8 Flash, PR Builder provides intelligent career guidance embedded directly in the authoring flow. API credentials remain securely isolated in backend serverless functions (/api/gemini).

- **Summary Generator:** Analyzes past job roles, skills, and target positions to draft executive summaries in under 2 seconds.
- **Bullet Point Optimizer:** Applies Google STAR methodology (Situation, Task, Action, Result) to transform generic descriptions into measurable accomplishments.
- **Skill Matcher:** Identifies high-frequency industry keywords missing from the candidate's profile based on the desired job role.

7. Security, Privacy & Database Architecture

Security and student data protection are deeply engineered into the platform:

- **Zero Sensitive Data Collection:** Resume forms strictly reject sensitive identifiers (no Aadhaar, SSN, PAN, full street addresses, or financial data).
- **Row-Level Security (RLS):** Every Supabase table (portfolios, resumes, domains, orders, entitlements) enforces authenticated RLS so students can only access their own data.
- **Frontend Bundle Hygiene:** Backend secrets (CASHFREE_SECRET_KEY, GEMINI_API_KEY) are never prefixed with VITE_ and are excluded from production builds.

8. Local Setup & Production Deployment Commands

Command	Execution Context	Description
npm run dev	kaizenq-identity	Launches local Vite development server on port 5173 with API proxies.
node scripts/build.js	Project Root	Compiles TypeScript, runs Vite bundle builder, and synchronizes dist/.
node scripts/run-all-tests.js	kaizenq-identity	Executes all 8 platform validation suites (33 Cashfree tests + security).
git push origin main	Git CLI	Triggers automated Vercel production deployment to https://prbuilder.xyz.

PRODUCTION DEPLOYMENT VERIFICATION

PR Builder is currently live at https://prbuilder.xyz. Subdomains route via *.prbuilder.xyz and direct portfolio fallbacks resolve at /p/[slug]. Cashfree payment mode is configured for live Indian settlement.